Irene Huang

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Education

Stony Brook University — Stony Brook, NY
M.B.A., General — *Expected Dec 2026*
GPA: 3.96/4.00

B.S., Computer Science — *May 2025*
GPA: 3.65/4.00

Research Interests

Quantitative decision-making for energy and infrastructure systems; optimization and multi-criteria decision analysis; renewable energy transition; resilient and sustainable infrastructure.

Research Experience

Text-Informed Nutrition Prioritization for West Africa — *2026*
Independent Research Project

- Developed a Python-based framework to identify foods that should be prioritized in resource-constrained settings by jointly considering nutrient completeness and nutrient-deficiency relevance.
- Retrieved PubMed literature through the ESearch/EFetch APIs and used text mining to quantify literature-based deficiency signals for individual nutrients.
- Applied TOPSIS to combine nutrient-completeness and literature-based deficiency signals into food rankings, identifying iron as the strongest deficiency signal and visualizing food scores and nutrient-deficiency rankings through heatmaps and comparative plots.

Data Envelopment Analysis for Nutritional Efficiency Modeling — *Jan 2026 – Aug 2026*
Research Assistant, Stony Brook University

- Used USDA nutrition data and data envelopment analysis (DEA) to evaluate the relative nutritional efficiency of foods based on beneficial nutrients and dietary risk factors, with applications to food selection in public health and hospital settings.
- Implemented optimization models across food groups using Excel, OpenSolver, VBA, and Python, calculating overall food efficiency scores and nutrient-level factor efficiencies to distinguish whether inefficiency was driven by the food as a whole or by specific nutrients relative to their targets.
- Conducted preliminary hierarchical clustering using centroid linkage on eligible food groups and generated dendrograms to identify groups of nutritionally similar foods, extending the efficiency analysis beyond individual scores and providing a basis for further comparative analysis.

Professional Experience

Stony Brook University — Stony Brook, NY
Marketing and Website Management Graduate Assistant — *Jun 2025 – Present*

- Build and maintain Graduate School webpages using HTML/CSS and JavaScript, implementing structural, accessibility, and content updates across institutional websites.
- Write and edit university news stories, program communication, and digital content, distilling complex academic and research information into clear materials for students, faculty, staff, and external audiences.

Renewable Energy Long Island — Brentwood, NY
Data Visualization Intern — *May 2026 – Aug 2026*

- Built a clean-energy dashboard for 15 Long Island municipalities, presenting data on renewable energy, green buildings, electric-vehicle infrastructure, and workforce development into accessible indicators for community stakeholders and local decision-makers.
- Cleaned, validated, and standardized municipal data from disparate public sources, identifying inconsistencies and creating a comparable dataset for cross-municipality comparison.
- Worked with project stakeholders to define performance indicators and design dashboard features aligned with

municipal reporting, comparison, and planning needs.

Vivvi Early Learning New York, NY
Admissions Intern *Jun 2022 – Aug 2022*

- Analyzed enrollment workflows and client data to identify process bottlenecks, contributing to a 39% increase in enrollment and improvements in the prospective-family onboarding process.
- Automated administrative workflows using Google Apps Script and maintained enrollment records for 240+ families, improving data consistency and reducing repetitive manual work.

Teaching Experience

Stony Brook University Stony Brook, NY

MBA 506: Leadership, Team Effectiveness, and Communications *Jan 2026 – May 2026*
Teaching Assistant

- Evaluated graduate-level written assignments and provided detailed feedback on argumentation, organization, evidence, and clarity, helping students strengthen both their analytical reasoning and professional writing.
- Assessed students' application of leadership, decision-making, and organizational behavior concepts while maintaining consistent grading standards across written coursework.

WSE 105: Opportunities in STEM and Beyond *Jan 2024 – May 2024*
Teaching Assistant

- Supported instruction and mentorship for students in the WISE Honors Program through classroom facilitation, academic support, grading, and individual student guidance.
- Collaborated with faculty to foster an inclusive learning environment designed to support academic success, community building, and participation in STEM.

AMS 210: Linear Algebra *Aug 2022 – Dec 2022*
Teaching Assistant

- Supported instruction for 170+ students in undergraduate linear algebra through weekly office hours, individualized problem-solving support, and responses to student questions.
- Developed grading rubrics and evaluated coursework to promote consistent assessment across a large-enrollment quantitative course.

Technical Skills

Programming	Python (Pandas, NumPy, scikit-learn, Matplotlib, PyTorch), R, SQL, Java, JavaScript, TypeScript, C
Quantitative Methods	Optimization, data envelopment analysis (DEA), multi-criteria decision analysis (MCDA), clustering, regression, time-series analysis, hypothesis testing, model validation
Web Development	HTML/CSS, React, Node.js, Express.js, Electron.js, Vite
Databases	MySQL, PostgreSQL, NoSQL, MongoDB
Tools	Excel, Google Sheets, Jupyter Notebook, VBA, OpenSolver, LaTeX, Git, VS Code